

ASP Parallel Planning

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1 Definitions

A STRIPS-like (multivalued) *planning task* according to blablabla is a 4-tuple $\langle F, s_0, s_*, \mathcal{O} \rangle$ with

- F is a finite set of *fluents* (variables that may change between states). Each $x \in F$ has an associated finite domain x^d of possible values.
- s_0 is the initial state, i.e. a total function that maps every $x \in F$ to a value in its domain x^d .
- s_* is a partial state specifying the goal conditions.
- $\mathcal{O}$ is a finite set of operators (actions) over F . Each action a has the form $a = \langle a^c, a^e \rangle$, where a^c and a^e are partial states denoting the precondition and postcondition of a .